

Joseph Zou

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EDUCATION

Columbia University

New York, NY

Master of Science in Artificial Intelligence, Concentration in AI and Advanced Computing Aug 2026 – Dec 2027

- **Relevant Coursework:** Neural Networks & Deep Learning, LLM-Based Generative AI Systems, Deep Learning on Edge, High-Performance Machine Learning, Machine Learning Theory, Reinforcement Learning

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science / Data Science, GPA: 3.91/4.0 Aug 2023 – May 2026

- **Relevant Coursework:** Artificial Intelligence, Data Mining & Machine Learning, Data Structures & Algorithms, Large-Scale Data Analytics, Statistical Programming & Data Management, Data Visualization

EXPERIENCE

Software Engineer Intern @ Arm

May 2026 – Aug 2026

- Profiling large language model inference on NVIDIA Jetson Orin Nano to measure tokens/sec, time-to-first-token (TTFT), latency, memory usage, and runtime behavior across edge AI deployment settings
- Running controlled LLM benchmarks across model variants, quantization settings, context lengths, and inference configurations to identify software bottlenecks in embedded inference pipelines
- Performance testing the software stack and developer ecosystem for a confidential Arm-based AI product, covering model frameworks, inference runtimes, libraries, APIs, deployment workflows, and developer tooling
- Creating performance evaluation platforms to compare how third-party AI software stacks run on Arm platforms

Data Science Intern @ John Deere

Aug 2025 – Dec 2025

- Trained Decision Tree, Random Forest, and XGBoost models on 1M+ records to forecast three-month demand across 1,000+ replacement parts
- Implemented recursive forecasting to generate realistic, sequential predictions without data leakage
- Engineered lag-based and temporal features to capture short-term demand patterns across 67-month time series
- Evaluated model performance using RMSE to guide improvements in forecast accuracy and business relevance

Data Science Intern @ BASF

Aug 2024 – May 2025

- Analyzed 10 years of California miticide application data to identify seasonal and regional usage patterns
- Estimated reapplication probabilities conditioned on spring active ingredient choice
- Conducted cost-benefit simulations to compare long-term outcomes of treatment strategies
- Performed geospatial county-level segmentation to inform region-specific product recommendations

PROJECTS

Edge AI Security Camera Assistant on Jetson Orin Nano

Jun 2026 – Present

- Built a fully local edge AI video pipeline on NVIDIA Jetson Orin Nano using a Logitech webcam, YOLO, TensorRT, and Objects365 for real-time person and object detection without cloud inference
- Integrated TensorRT-optimized object detection with class-name remapping, confidence filtering, and live OpenCV visualization to support low-latency embedded inference on an 8GB edge device
- Implemented timestamped event logging with SQLite and SSD-backed storage to record object appearances, disappearances, detection confidence, bounding boxes, and keyframe evidence
- Designed structured video memory for downstream LLM/VLM question answering over recent activity

Interpretable Generative Planning for Autonomous Driving

Jan 2026 – May 2026

- Collaborated in a capstone research team at Purdue University, advised by Prof. Ziran Wang and Dr. Jiaru Zhang, on diffusion-based autonomous driving
- Implemented bottleneck latent representations inspired by SODA to improve interpretability of generative trajectory planning models
- Added auxiliary objectives to encourage semantically meaningful latent representations of driving behavior
- Evaluated planning performance and representation quality on large-scale autonomous driving datasets using the NAVSIM simulation framework

TECHNICAL SKILLS

Languages: Python, R, SQL, SAS, Java, C/C++ **Tools:** PyTorch, scikit-learn, pandas, TensorRT, OpenCV, Git